

$$\int_0^{\sqrt{e^2-1}} \frac{x \ln(1+x^2)}{(1+x^2)^2} dx$$

$$u:=1+x^2 \quad du=2x dx$$

$$\frac{1}{2} \int_1^{e^2} \frac{\ln u}{u^2} du$$

$$w:=\ln u \quad dw=\frac{1}{u} du$$

$$\frac{1}{2} \int_0^2 \frac{w}{e^w} dw$$

$$y:=-w \quad dy=-dw$$

$$\frac{1}{2} \int_{-2}^0 ye^y dy$$

$$z:=y \quad dv=e^y dy$$

$$dz=dy \quad v=e^y$$

$$\frac{1}{2} ye^y \big|_{-2}^0 - \frac{1}{2} \int_{-2}^0 e^y dy$$

$$-e^{-2} - \frac{1}{2} e^y \big|_{-2}^0$$

$$-e^{-2} - \frac{1}{2} - \frac{1}{2e^2}$$

$$\frac{1}{2} - \frac{3}{2e^2}$$